CASE | Sales Team Competition

CASE EXERCISE
Inform a business decision using the two-sample t hypothesis test in a sales organization in a multi-national conglomerate.

1 Business Situation

A sales organization has two sales teams. Team #1 comprises 5 representatives and Team #2 comprises 4 representatives. Recently, Team #2 touted that its performance is better because the individual representatives on Team #2 performed better on average than those on Team #1 did.

2 Setup

Import necessary functions, define necessary functions, and configure other settings to ready the computing environment for analysis.

In [1]:
```
from numpy import *; from numpy.random import *; from datascience import *
from statsmodels.formula.api import *; from os.path import exists
f = ["../"*i + "setup.py" for i in range(10)]
exec(open(f[min([i for i, x in enumerate([exists(i) for i in f]) if x])]).read())
%matplotlib inline
```

3 Data

STEP 1

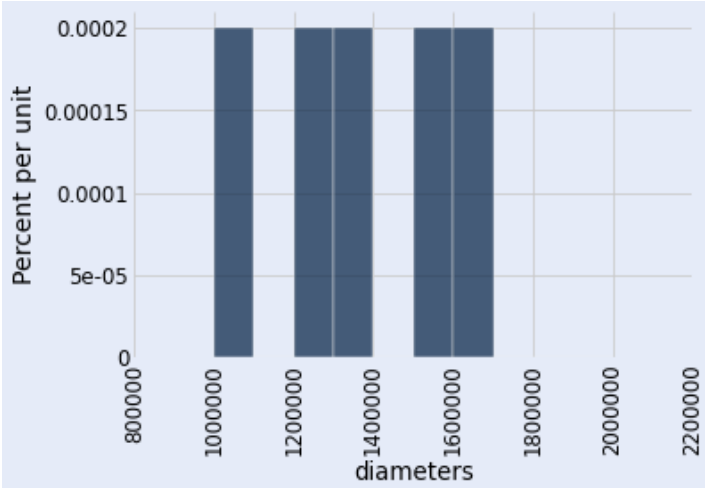
Sales Team #1 representatives makes sales of these amounts (in $):
1000000, 1500000, 1300000, 1200000, 1600000

For Sales Team #1, show the sample size, sample mean, sample standard deviation, and a histogram of the sampled sales (10 bins, range 1 million to 2 million).

In [3]:
```
sample1 = make_array(1000000, 1500000, 1300000, 1200000, 1600000)
sample1_size = len(sample1)
sample1_mean = mean(sample1)
sample1_std = std(sample1)

sample1_size, sample1_mean, sample1_std
Table().with_columns('diameters', sample1).hist(bins = 10, range = (1000000, 200000
```

Out[3]: (5, 1320000.0, 213541.56504062621)



STEP 2

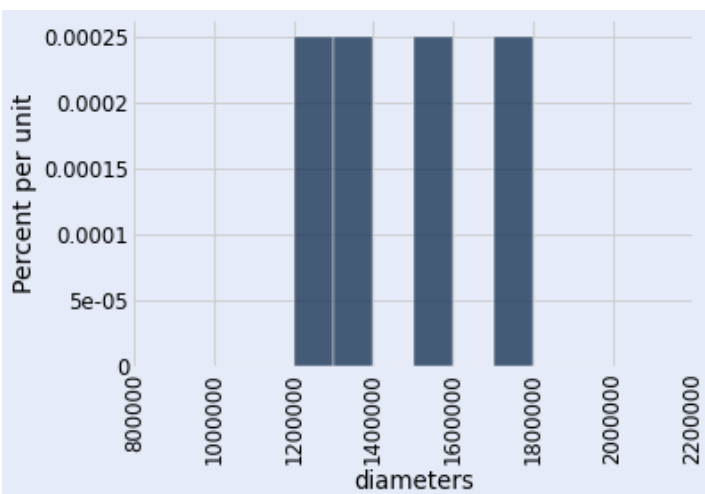
Sales Team #2 representatives make sales of these amounts (in $): 1300000, 1200000, 1700000, 1500000

For Sales Team #2, show the sample size, sample mean, sample standard deviation, and a histogram of the sampled sales (10 bins, range 1 million to 2 million).

In [4]:
```
sample2 = make_array(1300000, 1200000, 1700000, 1500000)
sample2_size = len(sample2)
sample2_mean = mean(sample2)
sample2_std = std(sample2)

sample2_size, sample2_mean, sample2_std
Table().with_columns('diameters', sample2).hist(bins = 10, range = (1000000, 200000
```

Out[4]: (4, 1425000.0, 192028.64369671521)



4 Analysis

STEP 1

Hypothesize that the difference between population mean sales is above or equal to 0. Or in other words, hypothesize that Sales Team #2 mean sales is not larger than Sales Team #1 mean sales. The alternative to this hypothesis is that the difference between mean sales is below 0.

In [5]:
```
pop_mean_diff_hypo = 0
```

STEP 2

Calculate the two-sample t.

In [6]:
```
two_sample_t = (sample1_mean - sample2_mean - pop_mean_diff_hypo) / sqrt((sample1_s
two_sample_t
```

Out[6]:
```
-0.775361859536156
```

STEP 3

Calculate the degrees of freedom for two samples.

In [7]:
```
ss1 = sample1_std**2/sample1_size
ss2 = sample2_std**2/sample2_size
over1 = 1/(sample1_size-1)
over2 = 1/(sample2_size-1)
df = (ss1+ss2)**2/(over1*ss1**2 + over2*ss2**2)
df
```

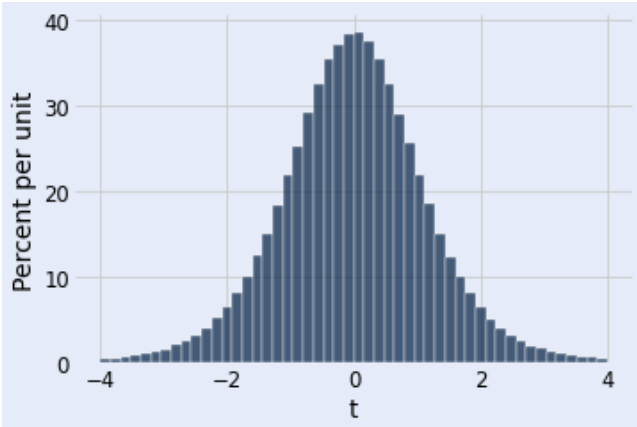
Out[7]:
```
6.846411092302966
```

STEP 4

Build a distribution of 1,000,000 values taken at random from the standard t distribution at the appropriate degrees of freedom. This is the reference distribution.
Show a few values from the reference distribution and a histogram of the reference distribution (50 bins, range -4 to 4).

In [8]:
```
dist = standard_t(df, 1000000)
dist
Table().with_column('t', dist).hist(bins = 50, range=(-4,4))
```

Out[8]:
```
array([-1.11489904, -0.0457072 , -0.13616271, ...,  0.70188791,
       -0.38119026, -0.79322412])
```



STEP 5

Calculate the probability that the difference between 2 new samples has a t value below or equal to the original two-sample t, assuming the difference in population means is above or equal to 0. This is the p-value.

In [11]:
```
p = count_nonzero(dist <= two_sample_t) / len(dist)
p
```

Out[11]:
```
0.231774
```

STEP 6

Calculate the critical value at significance level 0.05.
Hint: Note that the suspected difference between population means is BELOW the hypothetical zero difference between population means.

In [12]:
```
sig = .05
cv = percentile(sig*100, dist)
cv
```

Out[12]:
```
-1.9027452371036953
```

STEP 7

Conclude whether the hypothesis is likely correct at significance level 0.05.

In [13]:
```
p > sig
```

Out[13]:
```
True
```

5 Results

Record your analysis results.

- Team #1 reported making _____ sales. (count, not $)
- Team #1 reported mean sales of $_____.
- The sample standard deviation of Team #1's sales is $_____.
- Team #2 reported making _____ sales. (count, not $)
- Team #2 reported mean sales of $_____.
- The sample standard deviation of Team #1's sales is $_____.
- The two-sample t is _____.
- The degrees of freedom for two samples is _____.
- The p-value is _____.
- The critical value is _____.
- Should you conclude that the individuals on Team #2 perform better on average than the individuals on Team #1 do?
  - Yes. The high p-value indicates that Team #2 sales on average are statistically significantly higher than Team #1 sales on average.
  - Yes. The small two-sample t indicates that Team #2 sales on average are statistically significantly higher than Team #1 sales on average.
  - No, they've just been lucky. The large p-value indicates that Team #2 sales on average are not statistically significantly higher than Team #1 sales on average.
  - No, they've just been lucky. The small two-sample t indicates that Team #2 sales on average must be statistically significantly equivalent to Team #1 sales on average.
  - No, they've just been lucky. The negative critical value indicates that the Team #2 sales cannot exceed the significance level.

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